

Towards Machine Learning for Placement and Routing in Chip Design: a Methodological Overview

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Abstract

Placement and routing are two indispensable and challenging (NP-hard) tasks in modern chip design flows. Compared with traditional solvers using heuristics or expert-well-designed algorithms, machine learning has shown promising prospects by its data-driven nature, which can be of less reliance on knowledge and priors, and potentially more scalable by its advanced computational paradigms (e.g. deep networks with GPU acceleration). This survey starts with the introduction of basics of placement and routing, with a brief description on classic learning-free solvers. Then we present detailed review on recent advance in machine learning for placement and routing. Finally we discuss the challenges and opportunities for future research.

1 Introduction

The scale of integrated circuits (ICs) has increased dramatically, posing a challenge to the scalability of existing Electronic Design Automation (EDA) techniques and technologies. The increasing circuit density incurs additional issues for very large scale integration (VLSI) placers and routers as the feature size of modern VLSI design continues to drop and on-chip connectivity gets increasingly sophisticated. Due to increased on-chip connectivity, concentrated needs, and restricted resources, modern designs are prone to congestion issues and wire-length minimization, which has become a critical task at every stage of the design process. Accordingly, placement and routing – the elements of the design cycle physically arranges the locations and the courses of nets – becomes more crucial in modern VLSI.

Placement and routing are two of the most critical but time-consuming steps of the chip design process. Placement assigns various logic components like logic gates and functional blocks into discrete sites in the physical layout of a chip, and routing finishes the physical wiring of interconnections. The goal of placement and routing is to optimize power, performance, and area (PPA) metrics within constraints, e.g., placement density and routing congestion.

Placement can be regarded as a much more complicated variation of 2D bin packing problem with geometrical constraints. The latter is known to be NP-hard [Hartmanis, 1982]. The objective of placement is correlated to both the logical interconnection of the circuit design and the geometrical locations of logical components. As the quality of a placement solution cannot be accurately evaluated until routing, resulting in a long feedback loop in the design flow, modern placement needs to reduce routing congestion and improve routability in early stage. Routing has been proven to be an NP-hard problem [Hartmanis, 1982], even in the simplest case with only a few two-pin networks. Given the placement solution, it connects the pins of millions of networks with physical wires subjecting to the limited routing resources and complicated geometrical design rules. Routing is tightly coupled with placement and an excellent placement solution leads to better chip area utilization, timing performance, and routability.

There are also surveys on routing and placement. [Huang *et al.*, 2021] summaries a comprehensive review of existing ML studies for EDA field, most of which belongs to the four categories: decision making in traditional methods, performance prediction, black-box optimization, and automated design, ordered by increasing degree of automation. Although it covers various stages in the EDA flow, it provides shallow analysis for specific tasks, placement and routing, compared to ours. [Hamolia and Melnyk, 2021] introduces the ISPD 2015 dataset for comparison between classic methods and ML algorithms. [Rapp *et al.*, 2021] categorizes how ML may be used and is used for design-time/run-time optimization and exploration strategies of ICs, along with trends in the employed ML algorithms. [Markov *et al.*, 2015] reviews the history of placement research and the progress leading up to the state of the art. However, they only consider classic placer in the past 50 years, which is complementary to our survey which focuses on learning-based placer in recent years.

2 Problem Background and Classic Solvers

This section starts with the problem formulation and mainstream solvers which are traditionally learning-free. We then introduce recent applicable learning techniques.

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